Stat 266 – Unit 9 – Video Set 3 – Class Examples

Summary: Special results when sampling from a Normal population.

Small-Sample (*t-tools*) Confidence Interval for a Population Mean

Suppose we have a sample from a Normal population with unknown variance.

$\Rightarrow$ The two-sided $(1-\alpha)100\%\ C.I.$ for μ is $\boxed{\bar{Y} \pm t^*_{n-1,\alpha/2}\dfrac{S_Y}{\sqrt{n}}}$.

Pivot: $\dfrac{\bar{y}-\mu_y}{S_y/\sqrt{n}} \sim t_{n-1}$

Note: If the sample size is large this interval still applies and will be approximately the same as $\bar{Y} \pm z^*_{\alpha/2}\dfrac{S_Y}{\sqrt{n}}$.

$SD(\bar{y}) = \sigma_y/\sqrt{n}$ (unknown)

$SE(\bar{y}) = S_y/\sqrt{n}$

standard error

$, P(-t^*_{n-1,\alpha/2} < t_{n-1} < t^*_{n-1,\alpha/2}) = 1-\alpha$

Confidence Intervals for Variances:

Suppose you have independent samples from different populations.

- n_1 observations from $X \sim Normal(\mu_1, \sigma_1^2)$ and
- n_2 observations from $Y \sim Normal(\mu_2, \sigma_2^2)$

Pivot: $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

1. A $(1-\alpha)100\%$ confidence interval for σ^2 is $\left(\dfrac{(n-1)S^2}{\chi^{2^*}_{n-1,\alpha/2}}, \dfrac{(n-1)S^2}{\chi^{2^*}_{n-1,1-\alpha/2}}\right)$.

2. A $(1-\alpha)100\%$ confidence interval for $\dfrac{\sigma_2^2}{\sigma_1^2}$ is $\left(\dfrac{S_2^2}{S_1^2}F^{n_1-1^*}_{n_2-1,1-\alpha/2}, \dfrac{S_2^2}{S_1^2}F^{n_1-1^*}_{n_2-1,\alpha/2}\right)$.

Pivot: $\dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F^{n_1-1}_{n_2-1}$

Example A: An experiment to explore the pain thresholds to electrical shocks for males and females resulted in the data summary below. Suppose the pain thresholds for both males and females are normally distributed.

	Males	Females
n	14	10
Sample mean	16.2	14.9
Sample Variance	12.7	26.4

$\bar{y} \pm \{ t^*_{n-1, \alpha/2} \times S_y/\sqrt{n} \}$

Calculate (1 - α)100% confidence intervals for the means for males and females. $\Rightarrow$ 99% CI's

Males: C.V. $= t^*_{13, 0.005} = 3.012$

$\Rightarrow 16.2 \pm \{ 3.012 \times \sqrt{12.7}/\sqrt{14} \} \Rightarrow 16.2 \pm 2.86875$

$\Rightarrow (13.33, 19.07)$. $\Rightarrow$ a C.I. for μ_m

Note: The Z C.I. (C.V. $= Z^*_{0.005} = 2.576$).

$\Rightarrow 16.2 \pm \{ 2.576 \times \sqrt{12.7}/\sqrt{14} \} \Rightarrow \underbrace{(13.75, 18.65)}_{\text{narrower}}$.

Females: C.V. $= t^*_{9, 0.005} = 3.250$

$\Rightarrow 14.9 \pm \{ 3.250 \times \sqrt{26.4}/\sqrt{10} \} \Rightarrow 14.9 \pm 5.28062$

$\Rightarrow (9.62, 20.18)$. $\Rightarrow$ a C.I. for μ_F

Is there a real difference between μ_m and μ_F?

$\Rightarrow$ Find a C.I. for $\mu_M - \mu_F$.

Example B : Suppose you have two independent samples from different populations as follows:

- n_1 observations from $X \sim Normal(\mu_X, \sigma_X^2)$ and
- n_2 observations from $Y \sim Normal(\mu_Y, \sigma_Y^2)$.

Define a confidence interval for $\mu_X - \mu_Y$. Suppose σ_X^2 and σ_Y^2 are unknown.

Recall: If σ_X^2 and σ_Y^2 were known, then a $(1-\alpha)100\%$ C.I for $\mu_X - \mu_Y$ is

$$(\bar{X} - \bar{Y}) \pm Z^*_{\alpha/2} \times \sqrt{\frac{\sigma_X^2}{n_1} + \frac{\sigma_Y^2}{n_2}}.$$

Note: If n_1 and n_2 are large, then an approximate $(1-\alpha)100\%$ C.I. is

$$(\bar{X} - \bar{Y}) \pm Z^*_{\alpha/2} \times \sqrt{\frac{S_X^2}{n_1} + \frac{S_Y^2}{n_2}}$$ if σ_X^2 and σ_Y^2 are unknown.

Find an exact $(1-\alpha)100\%$ C.I. for $\mu_X - \mu_Y$ when sampling from Normal populations where σ_X^2 and σ_Y^2 are unknown:

1) $T = \dfrac{Z}{\sqrt{W/df}} \sim t_{df}$ where $Z \sim \text{Normal}(0,1)$, $W \sim \chi^2_{df}$ and Z and W are independent.

- $Z = \dfrac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\dfrac{\sigma_X^2}{n_1} + \dfrac{\sigma_Y^2}{n_2}}} \sim \text{Normal}(0,1).$

- $\dfrac{(n_1 - 1) S_X^2}{\sigma_X^2} \sim \chi^2_{n_1 - 1}$, $\dfrac{(n_2 - 1) S_Y^2}{\sigma_Y^2} \sim \chi^2_{n_2 - 1}$

$$W = \frac{(n_1 - 1) S_X^2}{\sigma_X^2} + \frac{(n_2 - 1) S_Y^2}{\sigma_Y^2} \sim \chi^2_{n_1 + n_2 - 2}$$

$$T = \frac{Z}{\sqrt{W/df}} = \frac{\dfrac{(\bar{X}-\bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\dfrac{\sigma_X^2}{n_1} + \dfrac{\sigma_Y^2}{n_2}}}}{\sqrt{\dfrac{\dfrac{(n_1-1)S_X^2}{\sigma_X^2} + \dfrac{(n_2-1)S_Y^2}{\sigma_Y^2}}{n_1+n_2-2}}} \sim t_{n_1+n_2-2}$$

Let's assume $\sigma_X^2 = \sigma_Y^2 = \sigma^2$.

$$\Rightarrow T = \frac{\dfrac{(\bar{X}-\bar{Y}) - (\mu_X - \mu_Y)}{\cancel{\sigma}\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}}{\dfrac{1}{\cancel{\sigma}}\sqrt{\underbrace{\dfrac{(n_1-1)S_X^2 + (n_2-1)S_Y^2}{n_1+n_2-2}}_{S_p^2}}} = \frac{(\bar{X}-\bar{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}$$

$$\sim t_{n_1+n_2-2}$$

$$S_p^2 = \frac{(n_1-1)S_X^2 + (n_2-1)S_Y^2}{n_1+n_2-2}$$

$\Rightarrow$ pooled estimate for σ^2,

$\Rightarrow$ a weighted average of S_X^2 and S_Y^2 weighted by their df.

$$1-\alpha = P\left(-t^*_{df,\alpha/2} < T < t^*_{df,\alpha/2}\right)$$

$$= P\left(-t^*_{df,\alpha/2} < \frac{(\bar{X}-\bar{Y})-(\mu_X-\mu_Y)}{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} < t^*_{df,\alpha/2}\right)$$

$$= P\left((\bar{X}-\bar{Y}) - t^*_{df,\alpha/2}\cdot S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}} < \mu_X-\mu_Y < (\bar{X}-\bar{Y}) + t^*_{df,\alpha/2}\cdot S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}\right).$$

So the $(1-\alpha)100\%$ C.I. for $\mu_X-\mu_Y$ is

$$(\bar{X}-\bar{Y}) \pm \left\{ t^*_{n_1+n_2-2} \times \underbrace{S_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}_{SE(\bar{X}-\bar{Y})} \right\}.$$

<u>Aside:</u> An approximate $(1-\alpha)100\%$ CI is

$(\bar{X}-\bar{Y}) \pm \left\{ t^*_{df}\sqrt{\frac{S_X^2}{n_1}+\frac{S_Y^2}{n_2}} \right\}$ without assuming equal variances.

(Stat 1x1 $\Rightarrow$ $df \simeq \min(n_1-1, n_2-1)$).

Example C: An experiment to explore the pain thresholds to electrical shocks for males and females resulted in the data summary below. Suppose the pain thresholds for both males and females are normally distributed.

	x Males	y Females
n	14	10
Sample mean	16.2	14.9
Sample Variance	12.7	26.4

Calculate a (1 - α)100% confidence interval for the difference in population means between males and females. $\Rightarrow$ 95%

Estimate: $\bar{x} - \bar{y} = 16.2 - 14.9 = 1.30$

$$S_p^2 = \frac{(n_1 - 1) S_x^2 + (n_2 - 1) S_y^2}{n_1 + n_2 - 2} = \frac{13(12.7) + 9(26.4)}{22} = 18.3045$$

$$SE(\bar{x} - \bar{y}) = S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = \sqrt{18.3045} \sqrt{\frac{1}{14} + \frac{1}{10}} = 1.7714$$

$$CV = t^*_{22, 0.025} = 2.074$$

Assuming $\sigma_x^2 = \sigma_y^2$ the 95% C.I. for $\mu_x - \mu_y$ is

$$\Rightarrow 1.30 \pm \{2.074 \times 1.7714\} \Rightarrow 1.30 \pm 3.6739$$

$$\Rightarrow (-2.374, 4.974).$$

Sampling from Normals where $\sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2$

<u>Linear Combinations of Means:</u> (k population means)

$$\theta = c_1\mu_1 + c_2\mu_2 + \cdots + c_k\mu_k$$

$$\hat{\theta} = c_1\bar{Y}_1 + c_2\bar{Y}_2 + \cdots + c_k\bar{Y}_k$$

$$\Rightarrow T = \frac{\hat{\theta} - \theta}{SE(\hat{\theta})} \sim t_{n-k}, \quad n = n_1 + \cdots + n_k$$

when $\sigma_1^2 = \cdots = \sigma_k^2$ and

$$SE(\hat{\theta}) = S_p\sqrt{\frac{c_1^2}{n_1} + \frac{c_2^2}{n_2} + \cdots + \frac{c_k^2}{n_k}} \quad \text{and}$$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2 + \cdots + (n_k-1)S_k^2}{n-k}.$$

The $(1-\alpha)100\%$ CI for θ

$$\hat{\theta} \pm \left\{ t^*_{n-k,\,\alpha/2} \times SE(\hat{\theta}) \right\}.$$

Example D: Suppose a process in under control if and only if $\sigma^2 < 0.36$. From a random sample of size 20 it was observed that $s^2 = 0.5$. Assuming that the observations are iid normal random variables, is the process is out of control? Answer the question by looking at confidence intervals for the true variance.

$\hookrightarrow$ Is $\sigma^2 > 0.36$?

A $(1-\alpha)100\%$ C.I. for σ^2 is

$$\left(\frac{(n-1)s^2}{\chi^{2*}_{n-1,\alpha/2}} , \frac{(n-1)s^2}{\chi^{2*}_{n-1,1-\alpha/2}} \right).$$

For 90% confidence $(\alpha = 0.10)$

$$\chi^{2*}_{19,0.05} = 30.144, \quad \chi^{2*}_{19,0.95} = 10.117.$$

The 90% CI for σ^2 is then

$$\left(\frac{19(0.50)}{30.144} , \frac{19(0.50)}{10.117} \right) \Rightarrow (0.315, 0.939).$$

Example E: An experiment to explore the pain thresholds to electrical shocks for males and females resulted in the data summary below. Suppose the pain thresholds for both males and females are normally distributed.

	X Males	Y Females
n	14	10
Sample mean	16.2	14.9
Sample Variance	12.7	26.4

Calculate a (1 - α)100% confidence interval for the ratio of population variances between males and females. $\Rightarrow 95\%$

The $(1-\alpha)100\%$ C.I. for σ_Y^2/σ_X^2 :

$$\left(\frac{S_Y^2}{S_X^2} \underbrace{F^{n_1-1\,*}_{n_2-2,\,1-\alpha/2}}_{F^{13\,*}_{9,\,0.975}=0.3019} , \frac{S_Y^2}{S_X^2} \underbrace{F^{n_1-1\,*}_{n_2-1,\,\alpha/2}}_{F^{13\,*}_{9,\,0.025}=3.8306} \right)$$

$$\Rightarrow \left(\frac{26.4}{12.7}(0.3019), \frac{26.4}{12.7}(3.8306) \right) \Rightarrow (0.6276, 7.9628)$$